



Multi-dimensional performance evaluation and influencing factors study of inter-chip thermal coupling in immersion-cooled heterogeneous multi-chip servers

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ABSTRACT

The rapid growth of AI computing demand is driving higher chip power density and a shift toward multi-chip server architectures to boost IT performance. The inter-chip thermal coupling characteristics are becoming increasingly significant, seriously affecting the heat dissipation performance of multi-chip immersion-cooled servers. However, it remains an open issue regarding its comprehensive thermal evaluation and influencing factors. This study developed a multi-dimensional thermal evaluation framework to characterize server thermal performance in terms of thermal safety, hotspot severity, temperature uniformity, and inter-chip thermal coupling strength. Based on this framework, the influencing factors assessment was conducted to quantify the effects of operating conditions (coolant inlet temperature and velocity) and structural designs (lateral chip offset and flow-guide baffle configurations) and to identify effective coupling-mitigation strategies. A three-dimensional numerical model of a heterogeneous multi-chip server was developed for heat dissipation performance analysis and it was first validated against experimental data, achieving a maximum deviation below 4.70 % in average chip temperature. The results indicate that inlet temperature mainly governs the absolute temperature level and hotspot severity with limited influence on thermal coupling, whereas increasing inlet velocity improves overall thermal performance with diminishing gains as a critical velocity is approached. Structurally, moderate lateral chip offset and properly designed baffles enhance bypass-coolant utilization and weaken inter-chip thermal coupling. Among all configurations, the 90 mm full-set baffle delivers the best overall enhancement, reducing the hotspot temperature, thermal non-uniformity coefficient, and downstream thermal resistance by 9.23 °C, 4.13, and 0.012 °C/W, respectively.

Nomenclature		Subscripts	
T	Temperature, °C	i	The rank of CPU
ρ	Density, kg/m ³	hotspot	Localized hotspot
c_p	Specific heat capacity, J/(kg·K)	rear	Average value of rear-row CPUs
$\bar{u}_i$	The velocity components on each axis, m/s	in	Immersion-coolant inlet conditions
u_t	The velocity pulsation value, m/s	Acronyms	
$\bar{p}$	Average pressure, Pa	AI	Artificial intelligence
λ	Heat conductivity, W/(m·K)	DC	Data center

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k	The turbulence kinetic energy, m ² /s ²	TDP	Thermal design power, W
ϵ	Turbulence kinetic energy dissipation rate, m ² /s ³	CPU	Central processing unit
G_k	Turbulent kinetic energy caused by velocity gradient, m ² /s ²	DIMM	Dual inline memory module
g_i	Gravitational acceleration component in the x_i direction, m/s ²	PCI	Peripheral component interconnect
μ_{eff}	Effective viscosity, Pa·s	PSU	Power supply unit

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β	The server temperature non-uniformity coefficient	HD	Hard disk
R	Thermal resistance, °C/W	3D	Three-dimension

1. Introduction

With the rapid growth of artificial intelligence (AI), cloud computing, and big data analytics, the global demand for high-performance computing continues to surge, driving the transition from traditional computing infrastructures to high-performance intelligent computing centers [1,2]. This trend is accompanied by the widespread deployment of high-power chips and ultra-dense AI clusters significantly increases both the scale and energy consumption of data centers (DCs) [3,4]. It is estimated that the growth rate of DCs will reach 12–14 % over the next five years [5]. By 2024, the electricity demand of DCs had risen to 400 TWh of electricity [6], with cooling systems accounting for nearly 40 % of the total energy consumption [7,8]. Meanwhile, the thermal design power (TDP) of next-generation AI processors has exceeded 300 W [9], with flagship chips such as NVIDIA's B200 reaching 1200 W. At the system level, the power demand of high-performance server racks has surpassed 30 kW, making conventional air-cooling approaches increasingly insufficient for managing such high heat fluxes [10,11]. Notably, more than half of electronic device failures are attributed to inadequate thermal management [12], underscoring the need to develop more efficient and energy-saving cooling technologies to support the sustainable evolution of DCs.

Compared with air cooling, liquid cooling technologies provide substantially higher heat-transfer capability and greater energy efficiency due to the superior thermal properties of liquid medias [13]. Existing liquid-cooling approaches mainly encompass cold-plate cooling, spray cooling, two-phase immersion cooling, and single-phase immersion cooling. Although cold-plate cooling offers low deployment cost, the solid-interface heat conduction adds unavoidable thermal resistance and limits the heat-transfer efficiency [14]. Spray cooling shows the advantages of low coolant usage and thermal contact resistance, but suffers from operational complexity and reliability issues stemming from frequent hot-swap procedures [15]. Immersion cooling directly immerses electronic devices in the dielectric coolant, further reducing contact thermal resistance and risk of leakage. Two-phase immersion systems exploit latent heat during boiling, achieving exceptional heat-removal capacity, but they introduce pressure oscillations and high maintenance demands [16,17]. Comparatively, single-phase immersion cooling employs thermally stable dielectric coolant, enabling efficient and stable heat removal with low system complexity, making it a promising option for future high-density intelligent-computing DCs.

Recent research on single-phase immersion cooling has primarily focused on liquid transport mechanisms, coolant formulation, operating-condition optimization, and local structural enhancement. Huang et al. [18,19] showed that pump-driven single-phase immersion cooling and anti-gravity flow deliver superior chip-level thermal performance and lower energy consumption compared with buoyancy-driven and gravity-assisted systems. Matsuoka et al. [20] demonstrated that the cooling performance of silicone oil, soybean oil, and fluorinated liquids decreases with increasing coolant viscosity. Chen et al. [21] further identified viscosity as the most influential thermo-physical property governing heat-transfer performance in fluorinated liquids, followed sequentially by thermal conductivity, density, and specific heat capacity. Moreover, both oil-based and fluorinated nano-fluids consistently exhibit superior cooling capability compared with their base liquids [22,23]. Chhetri et al. [24] examined the effects of key operating parameters on chip junction temperatures and reported that higher inlet temperature or power increases chip temperature, whereas increasing inlet velocity effectively suppresses temperature rise. Wang

et al. [25] demonstrated that increase the flow rate and inlet temperature of oil coolant within an appropriate operating range could enhance thermal transport capacity. Khoshvaght-Aliabadi et al. [26] used response surface methodology with a Box-Behnken design to systematically evaluated the influence of heat sink geometries and coolant operating conditions in a dual-CPU server. Li et al. [27] employed the response surface methodology to evaluate the influence of key server structural parameters, including heat-sink geometry, baffle height ratio, and outlet area. Shrigondekar et al. [28] investigated the effects of inlet-outlet configuration, fin spacing, and the clearance between the chip heat sink and the server chassis. Collectively, these studies have substantially advanced the understanding of immersion-cooling flow behavior, coolant thermophysical effects, and structural optimization strategies for improving server-level heat dissipation.

Despite these advances, existing studies have evaluated server thermal performance primarily from a thermal-safety-oriented or component-level perspective, typically using individual chip temperatures or local temperature distributions as the main criteria. This practice is effective for overheating prevention and baseline cooling design under simplified configurations. However, modern intelligent-computing servers are increasingly built on densely integrated heterogeneous multi-chip architectures with different power levels. Under immersion cooling, chips share the same coolant pathways, so upstream heat dissipation can preheat the coolant and alter internal flow distribution before it reaches downstream devices. Such coupled transport processes can create pronounced inter-chip thermal coupling and downstream hotspot risk that are not explicitly reflected by temperature-only evaluations. In addition, AI training and inference impose spatially and temporally heterogeneous power profiles across chips, which can further amplify coupling effects in operation. These considerations motivate a comprehensive thermal evaluation framework that explicitly quantifies inter-chip thermal coupling to support more reliable assessment and design of heterogeneous multi-chip servers.

To address these gaps, this study established a multi-dimensional thermal-evaluation framework for immersion-cooled heterogeneous multi-chip servers, with explicit consideration of inter-chip thermal coupling. Based on this framework, operating conditions and internal structural configurations were systematically examined to identify effective coupling-mitigation measures and extract design-relevant trends for server-level thermal management. A validated three-dimensional (3D) heterogeneous multi-chip server model representative of AI computing nodes was developed to support the evaluation and mechanism interpretation. The main contributions are summarized as follows.

- (i) A server-level, multi-metric evaluation scheme is established to explicitly quantify inter-chip thermal coupling, together with thermal safety, hotspot severity, and temperature non-uniformity.
- (ii) The formation and impact of thermal coupling phenomena in immersion-cooled multi-chip servers are clarified through combined thermal and flow-field analyses.
- (iii) The distinct roles and governing mechanisms of operating conditions and structural configurations in regulating server thermal performance are systematically identified.

2. Methodology

2.1. Framework for multi-dimensional evaluation and optimization of inter-chip thermal coupling

Fig. 1 presents the overall framework for multi-dimensional performance evaluation and influencing factors analysis of inter-chip thermal coupling for the immersion-cooled heterogeneous multi-chip server. A representative server configuration with non-uniform chip power levels

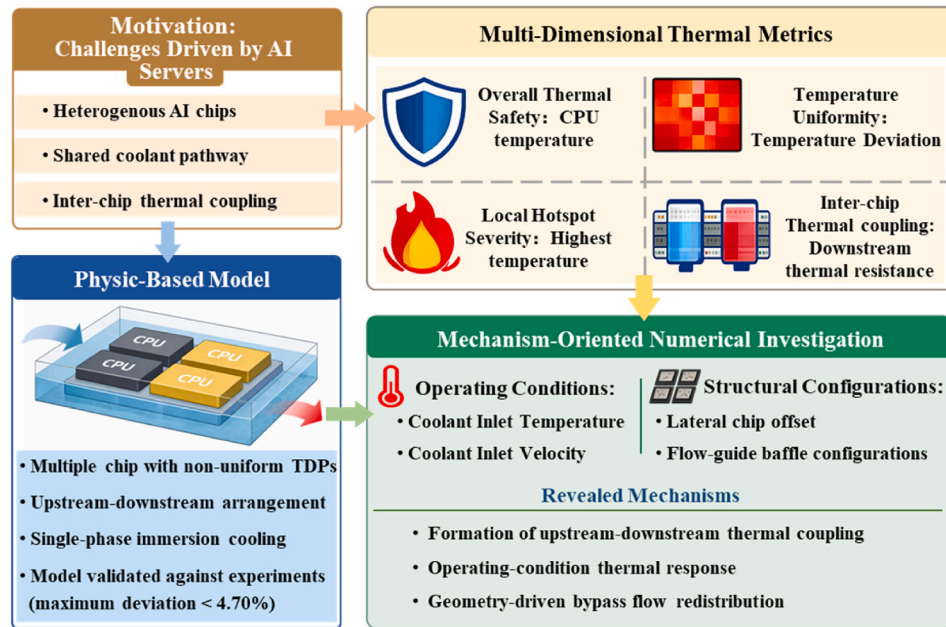


Fig. 1. Framework for mechanism-oriented thermal coupling analysis of a heterogeneous multi-chip server.

arranged along the coolant flow direction was established to capture upstream-downstream interactions under shared coolant pathways. The thermal performance was comprehensively evaluated using a coupling-aware metric set that jointly describes overall thermal safety (chip temperature level), hotspot severity (localized hotspot temperature), temperature uniformity (temperature deviation), and coupling strength quantified by the downstream thermal resistance.

Based on the above evaluation scheme, operating parameters (coolant inlet temperature and inlet velocity) and structural variables (lateral chip offset and flow-guide baffle configurations) were systematically explored to identify operating conditions and layouts that mitigate coupling and improve server-level performance. The resulting flow and temperature fields were analyzed to interpret the governing mechanisms, emphasizing the formation of thermal coupling, operating-parameter sensitivity of the server thermal response, and bypass-flow redistribution induced by chip offset and baffles. Overall, the proposed framework enables a systematic evaluation of design-relevant trends and actionable guidance for mitigating inter-chip thermal coupling in immersion-cooled heterogeneous multi-chip servers.

2.2. Thermal-performance indicators

To operationalize the framework introduced in Section 2.1, four complementary metrics were defined to evaluate the thermal performance of the heterogeneous multi-chip server under single-phase immersion cooling. Their definitions and calculation procedures are summarized in Table 1. In the subsequent sections, these indicators were employed to quantify server-level performance and thermal coupling mitigation effects under varying operating conditions and component arrangements.

2.3. Numerical model

2.3.1. Governing equations

To simplify the numerical modeling, the following assumptions were adopted: (i) the coolant flow inside the server is treated as three-dimensional, steady, and incompressible, with constant thermophysical properties assumed owing to the limited temperature range (15–45 °C); (ii) the flow regime is assumed to be turbulent, considering the strong flow disturbances, separation, and recirculation within the coolant induced by the complex internal geometry of the server; (iii) thermal radiation is neglected because convective heat transfer

Table 1
Definitions of the evaluation indicators.

Index	Chip temperature (T_{CPUi})	Localized hotspot temperature ($T_{hotspot}$)	Server thermal non-uniformity coefficient (β)	Downstream chip thermal resistance (R_{rear})
Significance	A primary indicator reflecting overall thermal safety by constraining chip temperatures within acceptable limits.	An essential metric for identifying hotspot severity and assessing local thermal risks inside the server.	A global metric describing the uniformity of the temperature field, closely related to thermal stress.	A key indicator for quantifying the intensity of inter-chip thermal coupling between upstream and downstream chips.
Feature	Represents the average operating temperature of each chip.	Represents the highest temperature observed among all chip.	Describes the deviation of local temperatures from the server-level mean temperature.	Reflects how significantly the downstream chips are affected by the heated coolant from upstream devices.
Formula	Monitoring data.	Monitoring data.	$\beta = \frac{\sum \left(T_{CPUi} - \frac{1}{N} \sum T_{CPUi} \right)^2}{N - 1}$	$R_{rear} = \frac{1}{2} \left(\frac{T_{CPU3} - T_{in}}{P_{CPU3}} + \frac{T_{CPU4} - T_{in}}{P_{CPU4}} \right)$
Objective	Maintain chip temperatures below the safe limit to prevent reliability degradation.	Minimize $T_{hotspot}$ to avoid local overheating and ensure stable operation under high power density.	Minimize β to reduce temperature gradients and mitigate thermal-induced stress.	Minimize R_{rear} to weaken thermal coupling and enhance the cooling effectiveness for downstream chips.

dominates under liquid immersion and the temperature levels are moderate; (iv) ambient heat losses and viscous dissipation are ignored, as their contributions are negligible compared with the internal convective heat transfer within the server enclosure. The RNG k - ϵ turbulence model was selected to describe the turbulent flow, as it offers improved accuracy for rapidly strained flows while maintaining relatively low computational cost [29]. The governing Reynolds-averaged Navier-Stokes equations used in this study [30,31] are given as follows.

Continuity equation:

$$\frac{\partial(\rho \bar{u}_i)}{\partial x_i} = 0 \quad (1)$$

Momentum equation:

$$\frac{\partial}{\partial x_j} (\rho \bar{u}_i \bar{u}_j) = -\frac{\partial \bar{p}}{\partial x_i} + \mu \frac{\partial^2 \bar{u}_i}{\partial x_j^2} + \rho g_i + \frac{\partial}{\partial x_j} (-\rho \bar{u}_i \bar{u}_j') \quad (2)$$

Energy equation:

for coolant region

$$\frac{\partial}{\partial x_i} (\rho c_p \bar{u}_i \bar{T}) = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial \bar{T}}{\partial x_i} - \rho c_p \bar{u}_i \bar{T}' \right) \quad (3)$$

for solid region

$$\lambda \frac{\partial}{\partial x_i} \left(\frac{\partial \bar{T}}{\partial x_i} \right) = 0 \quad (4)$$

k -equation:

$$\frac{\partial}{\partial x_i} (\rho k \bar{u}_i) = \frac{\partial}{\partial x_j} \left(\alpha_k \mu_{\text{eff}} \frac{\partial k}{\partial x_j} \right) + G_k - \rho \epsilon \quad (5)$$

ϵ -equation:

$$\frac{\partial}{\partial x_i} (\rho \epsilon \bar{u}_i) = \frac{\partial}{\partial x_j} \left(\alpha_\epsilon \mu_{\text{eff}} \frac{\partial \epsilon}{\partial x_j} \right) + C_{1\epsilon} G_k \frac{\epsilon}{k} - C_{2\epsilon} \rho \frac{\epsilon^2}{k} - R_\epsilon \quad (6)$$

R_ϵ can be calculated by

$$R_\epsilon = \frac{C_\mu \rho \eta^3 \left(1 - \frac{\eta}{\eta_0} \right)}{1 + \gamma \eta^3} \frac{\epsilon^3}{k} \quad (7)$$

where $\bar{u}$ denotes the velocity components of the coolant in each direction; $\bar{T}$ and $\bar{p}$ are the average temperature and pressure; $\bar{u}'$ and $\bar{T}'$ correspond to the velocity and temperature fluctuations, respectively. The symbols ρ , λ , c_p and μ represent the density, heat conductivity, specific heat and dynamic viscosity of the immersion coolant. k and ϵ indicate the turbulence kinetic energy and dissipation rate, while G_k is the turbulent kinetic energy due to the mean velocity gradients. μ_{eff} is the effective kinetic viscosity, and η denotes the ratio of turbulence to mean strain time. The remaining model constants are given as $C_{1\epsilon} = 1.42$, $C_{2\epsilon} = 1.68$, $C_\mu = 0.0845$, $\gamma = 0.012$ and $\eta_0 = 4.38$.

2.3.2. Boundary condition

In the simulation, all heat-generating components were modeled as

uniform volumetric heat sources. The thermophysical properties of the key components and the immersion coolant are summarized in Table 2. The server inlets were specified as velocity inlets, and the outlet was defined as a pressure outlet. The interfaces between the electronic components and the coolant were treated as fluid-solid coupling surfaces, with the solid surfaces being non-slip. The server walls were assumed adiabatic to neglect heat losses to the environment.

2.3.3. Numerical software

The flow and heat transfer within the multi-chip server were simulated using ANSYS 2023R1. The server model was discretized with an unstructured mesh generated by Fluent Meshing, and local refinement was applied to critical components, including CPUs and heatsinks. The Coupled algorithm was employed to resolve the pressure-velocity coupling. And the PRESTO! scheme, suitable for high-rotation and rapidly varying pressure flows [21], was applied for pressure correction. Momentum and energy equations were discretized using a second-order upwind scheme. Sub-relaxation factors of 0.3, 0.5, and 0.75 were applied to pressure, momentum, and energy, respectively. Convergence was considered achieved when residuals reached 10^{-6} for energy and 10^{-3} for all other equations, and the monitored CPU temperatures remained stable.

3. Case study

3.1. Server geometry and component configuration

To examine inter-chip thermal coupling in a reproducible setting, a representative immersion-cooled server was selected as the baseline case study. The geometry was refined from the configuration reported by Wang et al. [32], while retaining the key flow passages and component layout that govern internal coolant distribution. The overall geometry and internal component arrangement are illustrated in Fig. 2 (a). The server chassis measures 500 mm in length, 100 mm in width, and 850 mm in height. Two liquid inlets are located at the bottom of the chassis, whereas the outlet is positioned along the upper sidewall. Inside the server, the electronic components comprise the central processing unit (CPU), dual inline memory module (DIMM), peripheral component interconnect (PCI), power supply unit (PSU), and hard disk (HD). As shown in Fig. 2 (b), finned heatsinks are mounted on the CPUs to strengthen convective heat transfer between the chips and the surrounding immersion coolant. The heat sinks feature a base thickness of 7 mm, a height of 42 mm for the 14 fins, a longitudinal dimension of 90 mm, and a fin spacing of 3 mm. Table 3 summarizes the dimensions, the TDP and quantity of each component. Notably, the high-TDP chips (CPU 1 and 2, 900 W) are arranged in the upstream region to ensure preferential cooling, whereas the low-TDP chips (CPU 3 and 4, 600 W) are positioned downstream. This layout exposes the downstream chips to pre-heated coolant, intentionally amplifying inter-chip thermal coupling and providing a clearer basis for subsequent investigation of thermal interaction.

3.2. Grid-independence test and model validation

Considering the computational efficiency and accuracy, grid-independence verification was performed. In this study, nine different grid systems were tested. The coolant inlet temperature and flow velocity were set to 25 °C and 0.25 m/s, respectively. The average CPU temperatures for each grid system are shown in Fig. 3. As the cell count increases from 946,891 to 6,448,567, the average CPU temperature decreases. However, beyond 7,342,399 cells, the temperature variation between consecutive grid sets remains below 0.5 %. Therefore, a grid system with 7,342,399 cells was chosen for the subsequent numerical simulations.

To further validate the proposed simulation model, a comparative study was conducted based on the single-phase immersion cooling

Table 2
Thermophysical properties of key components and the coolant.

Coolant and component	Material	Density (kg/m ³)	Specific thermal capacity (J/(kg·K))	Thermal conductivity (W/(m·K))	Dynamic viscosity (kg/(m·s))
CPU	Silicon	2329.5	697.28	153.04	—
PCB	Fr-4	1200	1310	0.35	—
Heat sink	Copper	8978	381	387.6	—
Coolant	FC-40 (25 °C)	1870	1050	0.067	0.0041

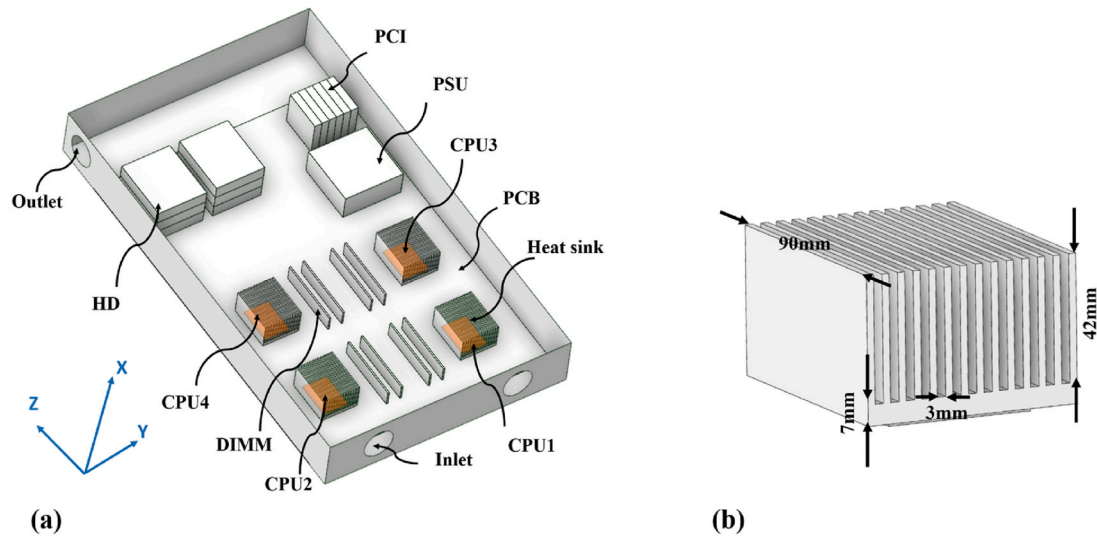


Fig. 2. Geometry of the heterogeneous server: (a) Arrangement of components (b) Structure of heat sink.

Table 3

The dimensions, the TDP and quantity of each component.

Component name	Geometrical dimension (mm)	Thermal design power (W)	Quantity
CPU	$70 \times 50 \times 3$	900/600	4
PCB	$700 \times 500 \times 2$	0	1
PCI	$80 \times 15 \times 70$	10	6
HD	$120 \times 100 \times 25$	10	6
PSU	$120 \times 120 \times 50$	100	1
DIMM	$130 \times 3 \times 30$	10	8

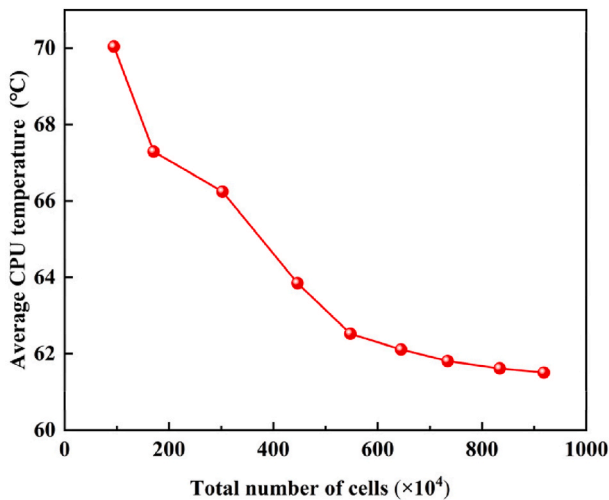


Fig. 3. Grid-independence analysis.

experiments of Luo et al. [22]. The experimental configuration was geometrically simplified for numerical implementation, as illustrated in Fig. 4 (a). The comparisons between experimental measurements and numerical results under inlet conditions of 25 °C and mineral oil flow velocities ranging from 0.04 to 0.32 m/s are presented in Fig. 4 (b). The average temperature of the heat source exhibits good agreement with the experimental data over the investigated velocity range, with relative deviations between 0.14 % and 4.70 %. The observed discrepancies are mainly attributed to experimental measurement uncertainty and geometric simplifications adopted in the numerical model. Overall, the results indicate that the proposed numerical model provides reliable

predictions under the examined operating conditions.

4. Results and discussion

This section focuses on the inter-chip thermal coupling behavior of an immersion-cooled heterogeneous multi-chip server under varying operating conditions and internal layouts. Based on the thermal evaluation metrics defined in Table 1, the influences of coolant inlet temperature and inlet velocity were first analyzed to distinguish variations in absolute temperature levels from changes in coupling strength along the flow path. Structural modifications, including lateral chip offset and flow-guide baffles, were then examined as thermal coupling mitigation approaches through bypass-flow redistribution. The corresponding temperature and flow fields were analyzed to interpret the governing physical mechanisms, encompassing coolant preheating, velocity-dependent convective behavior, and geometry-induced flow redistribution. Together, these analyses establish a consistent framework linking operating conditions, structural design, and inter-chip thermal coupling at the server level.

4.1. Effect of coolant inlet temperature

The influence of coolant inlet temperature on server thermal behavior was examined under a fixed inlet velocity of 0.25 m/s, with the inlet temperature varied from 15 to 45 °C in 5 °C increments. As depicted in Fig. 5 (a), CPU temperatures and localized hotspot temperature increase nearly linearly with inlet temperature, consistent with the trend reported by Gong et al. [33]. Specifically, each 5 °C increase in inlet temperature results in an approximately equal rise in chip temperatures.

Notably, despite the higher TDP of the upstream chips, the downstream chips are approximately 13.35 °C hotter than the upstream chips, and the thermal hotspot is consistently situated downstream. To further clarify this upstream-downstream temperature difference, Fig. 6 presents the temperature and velocity fields of the multi-chip server under an inlet temperature of 25 °C. A distinct contrast is observed: the upstream chips exhibit downstream-oriented high-temperature zone, whereas the downstream chips develop a pronounced upstream-side thermal accumulation. This is attributed to the upstream chips being cooled by the incoming coolant at its initial temperature under more favorable local flow conditions. In contrast, the downstream chips are exposed to coolant that has been preheated by the upstream devices, which diminishes local heat-removal capability and promotes the formation of downstream hotspots. These findings demonstrate

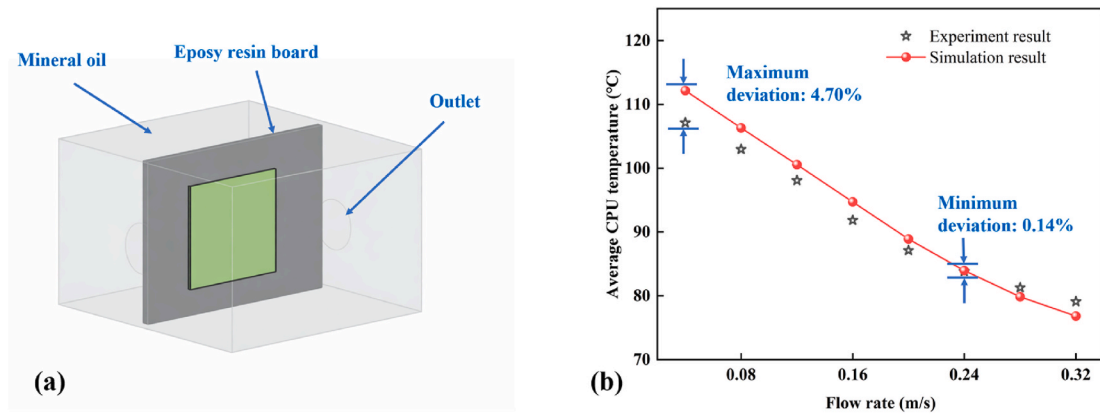


Fig. 4. Validation of the server model. (a) Validation model (b) Comparison of results.

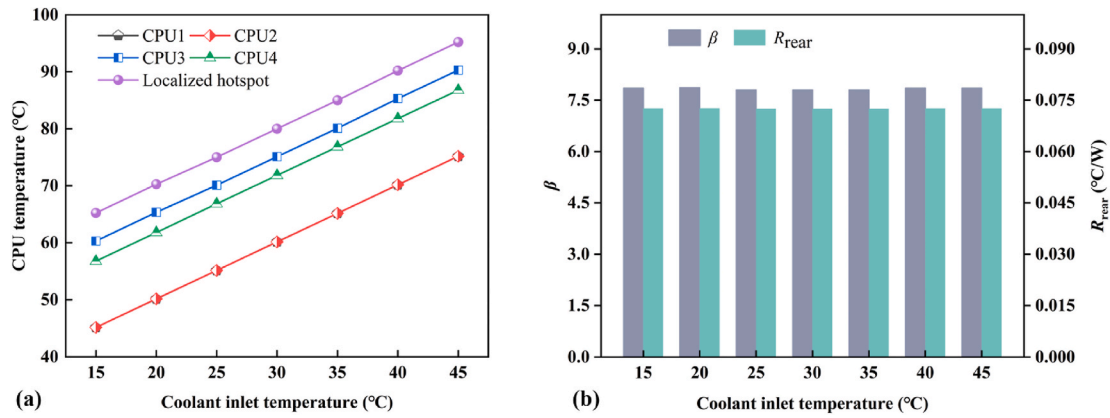


Fig. 5. Effect of coolant inlet temperature on server-level thermal performance.

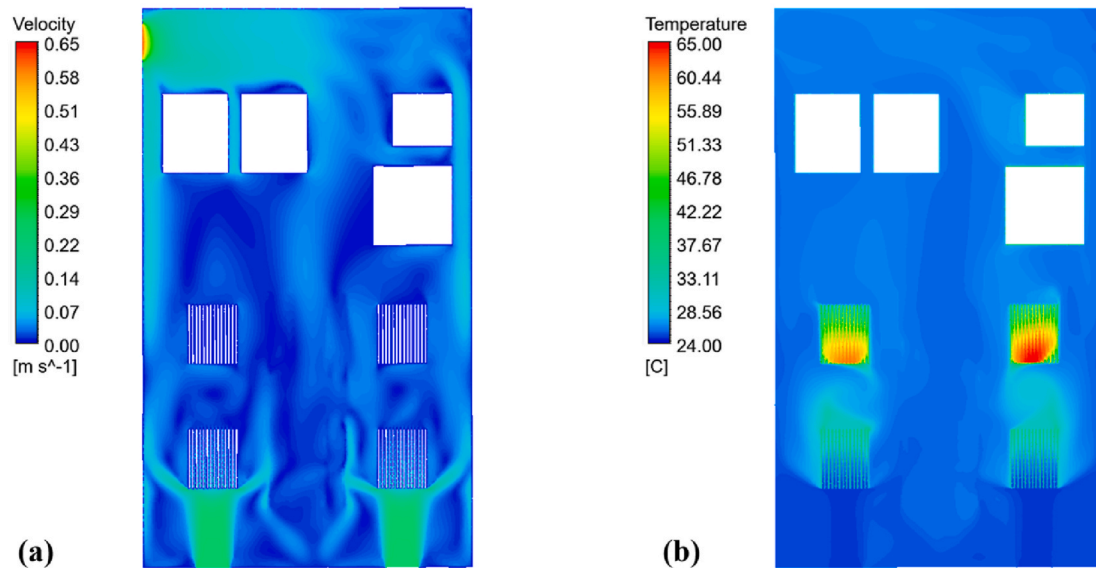


Fig. 6. Velocity and temperature fields within the server ($X = 35$ mm). (a) Velocity field (b) Temperature field.

pronounced inter-chip thermal coupling, where chip temperatures are governed jointly by power levels and their spatial arrangement along the flow path. Accordingly, the role of flow redistribution induced by structural layout warrants further examination to improve thermal performance.

Given the nearly linear dependence of chip temperatures on the inlet coolant temperature, both the thermal non-uniformity coefficient and the downstream chip thermal resistance remain essentially constant, with values of 7.83 and 0.072 °C/W, respectively (Fig. 5 (b)). The invariance of these metrics indicates that adjusting the inlet temperature

primarily alters the absolute temperature level without modifying the internal flow redistribution. Consequently, lowering the inlet temperature effectively suppresses the magnitude of local temperature peaks, but does not improve the overall temperature uniformity or mitigate the inter-chip thermal coupling.

4.2. Effect of coolant inlet velocity

To investigate the influence of coolant inlet velocity on server thermal management performance, the inlet velocity was varied from 0.05 to 0.35 m/s in increments of 0.05 m/s, and the inlet temperature was maintained at 25 °C. The resulting variations in key thermal performance metrics are presented in Fig. 7. The chip temperatures, server thermal non-uniformity coefficient, and downstream chip thermal resistance all decrease with increasing inlet velocity, but the rate of improvement diminishes progressively. In the low-velocity regime (0.05–0.15 m/s), the local hotspot temperature decreases by 97.76 °C, the thermal non-uniformity coefficient drops by 36.73, and the downstream chip thermal resistance decreases by 0.168 °C/W. By contrast, in the higher-velocity range (0.25–0.35 m/s), the corresponding reductions are only 10.28 °C, 4.16, and 0.014 °C/W. These results suggest the presence of a critical inlet velocity beyond which further increases in coolant flow rate provide only marginal thermal benefits, consistent with earlier observations in the literature [19,32].

A distinct difference is observed in the thermal response of the two chip rows. The downstream chips show a substantially stronger temperature reduction over the examined velocity range, whereas the upstream chips show only modest variation. Once the inlet velocity exceeds about 0.2 m/s, the upstream chip temperatures remain essentially unchanged, while the downstream chips continue to experience measurable cooling. This behavior arises from the markedly different flow environments experienced by the two chip rows. The upstream chips are exposed to relatively high local velocities, causing the enhancement in convective heat transfer to reach saturation at a comparatively low inlet velocity. However, the downstream chips receive reheated coolant and operate under lower local velocities imposed by upstream flow resistance. Thus, the downstream chips exhibit a higher critical velocity and a more pronounced thermal response to changes in inlet velocity, while the upstream row reaches saturation first and consequently sets the server-level critical velocity.

Within the effective velocity-regulation range, increasing the inlet velocity can enhance convective heat transfer, thereby attenuating the thermal coupling between upstream and downstream chips and yielding a more uniform temperature distribution across the server. However, as the inlet velocity approaches the critical threshold governed by the upstream chips, further gains in server-level heat dissipation become minimal, whereas the pumping power increases continuously. Under these conditions, adjusting the inlet temperature may provide a more

energy-effective pathway for further optimizing server thermal management.

4.3. Effect of chip layout

Previous analysis has confirmed the thermal performance of chips is affected by coolant operating conditions and their geometric layout. Building on this, this subsection investigates the influence of chip arrangement on server thermal behavior. Considering that the longitudinal positions of components are largely predetermined by the motherboard architecture, this study concentrates on lateral offset as the primary chip layout variable. The upstream chip positions remain fixed, while the downstream chips are incrementally shifted toward the chassis sidewall. Under inlet conditions of 0.25 m/s and 25 °C, a series of offset ratios (defined as the lateral displacement normalized by chip width) was evaluated to quantify its impact on temperature distribution, thermal uniformity, and inter-chip thermal coupling.

Fig. 8 presents the influence of lateral chip offset on the thermal performance of the multi-chip server. The results demonstrate that chip arrangement exerts a pronounced effect on both local and system-level thermal behavior. Within the lateral offset range of 0–75 %, the downstream chip temperature, server thermal non-uniformity coefficient, and downstream chip thermal resistance decrease monotonically as the offset increases. For instance, the temperature of the warmer downstream chip (CPU3) decreases from 70.09 °C to 63.94 °C, while the hotspot temperature drops from 74.99 °C to 68.08 °C. Correspondingly, the thermal non-uniformity coefficient reduces from 7.80 to 4.72, and the average thermal resistance declines of downstream chips from 0.072 °C/W to 0.064 °C/W. When the offset ratio increases further from 75 % to 100 %, a slight deterioration of the thermal metrics is observed (CPU3 temperature of 64.43 °C, hotspot temperature of 68.70 °C, non-uniformity coefficient of 5.20, and thermal resistance of 0.065 °C/W). Nevertheless, the performance remains substantially better than the baseline configuration. These trends indicate that moderate lateral offset effectively suppresses hotspot formation, enhances temperature uniformity, and weakens inter-chip thermal coupling. Meanwhile, a slight increase in pressure drop is observed with increasing lateral offset. The pressure drop increases modestly from 368.34 Pa to 368.72 Pa as the offset ratio varies from 0 % to 100 %, implying that the thermal enhancement is accompanied by a minor hydraulic penalty.

To reveal the mechanisms underlying the observed thermal improvements, the coolant flow fields in the baseline and offset configurations are examined. In the baseline layout (Fig. 6 (a)), the coolant enters the server cavity and first encounters the upstream chips. The upstream components induce substantial geometric blockage, causing part of the incoming flow to bypass the primary heat-exchange region and divert into the lateral channels. This bypassed flow subsequently mixes with the heated effluent from the upstream chips before reaching

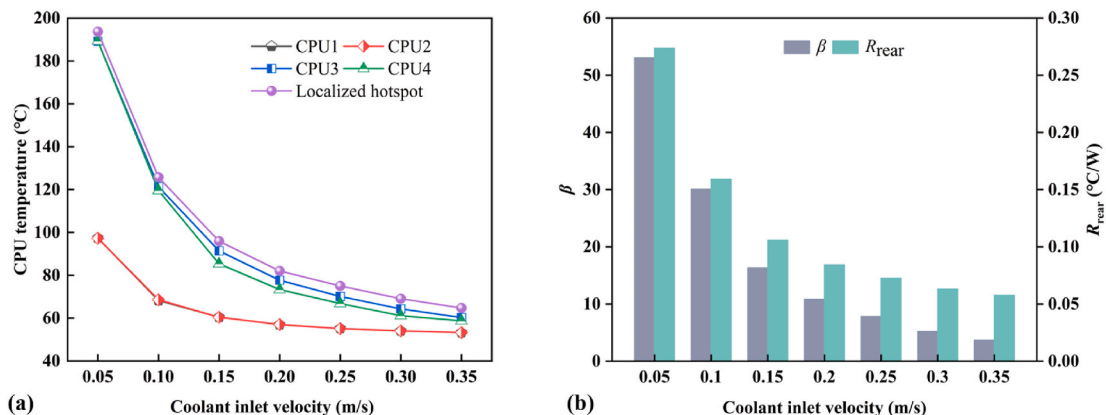


Fig. 7. Effect of coolant inlet velocity on server-level thermal performance.

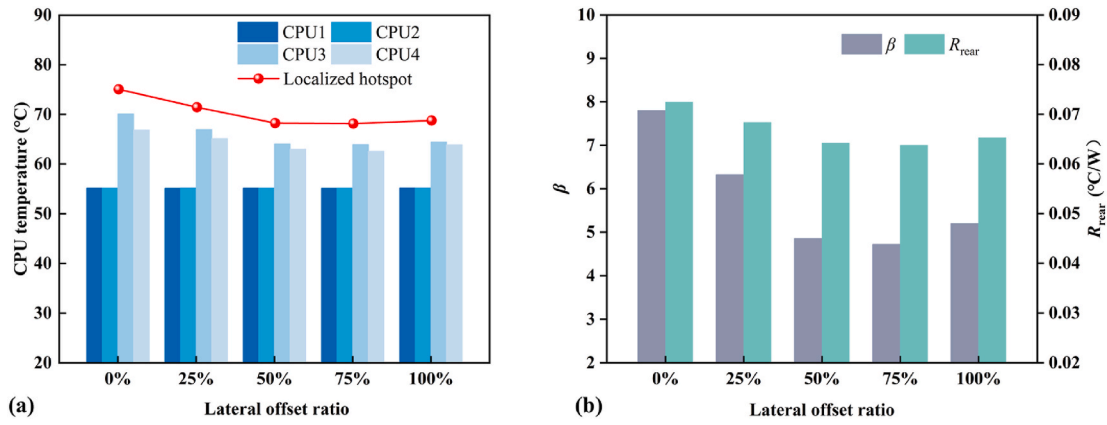


Fig. 8. Effect of lateral chip offset on server-level thermal performance.

the downstream row. The resulting mixture exhibits elevated temperature and reduced local velocity, giving rise to the higher temperatures and thermal resistance previously noted. As shown in Fig. 9, when the set ratio increases from 25 % to 75 %, the downstream chips progressively detach from the primary wake generated by the upstream row and gain more direct exposure to the cooler, higher-velocity bypass streams. As a result, the local coolant velocity around the downstream chips increases and the influence of reheated fluid diminishes, yielding a substantial improvement in their convective heat removal. However, at an offset ratio of 100 %, the downstream chips become positioned too close to the sidewall, which inhibits the effective access of bypass coolant and limits further enhancement of heat transfer. This geometric constraint accounts for the slight decline in thermal performance observed at the highest offset setting.

In conclusion, a moderate lateral offset provides a robust pathway to improve temperature uniformity and attenuate inter-chip thermal coupling in multi-chip servers. However, excessive offset reduces the available flow passage and limits coolant access, thereby diminishing the attainable heat-transfer enhancement. This highlights the need for a balanced offset configuration that maximizes cooling benefits without reducing the effective coolant supply to the downstream chips.

4.4. Effect of flow-guide baffle

The previous section demonstrated that lateral chip offset enhances the utilization of bypass coolant and substantially weakens inter-chip thermal coupling in multi-chip servers. Motivated by this mechanism, an alternative approach was explored in which the bypass flow was guided through installed baffles to further regulate coolant distribution.

Three flow-guide baffle placements were examined, namely front-set, rear-set and full-set baffles (Fig. 10). In addition, baffle heights of 60 mm, 75 mm and 90 mm were selected to ensure a significant influence on the internal flow field. These configurations enable a clear characterization of the combined effect of baffle position and height on heat-dissipation performance.

Table 4 summarizes the server thermal performance under different baffle configurations at an inlet velocity of 0.25 m/s and an inlet temperature of 25 °C. In all cases, the upstream chip temperatures differ from the baseline by less than 0.3 °C, indicating that the baffles exert only a negligible effect on upstream thermal conditions. By contrast, the downstream chips show a stronger sensitivity to baffle arrangement. Among the three placements, the front-set baffle consistently improves downstream chip cooling and enhances server-level temperature uniformity. The rear-set baffle leads to a deterioration in thermal performance across nearly all heights, whereas the full-set baffle exhibits a noticeable adverse effect only at the lowest height (60 mm). For the 60-mm baffle height, the front-set baffle lowers the hotspot temperature from 74.99 °C to 73.36 °C, reduces the thermal non-uniformity coefficient from 7.80 to 7.09, and decreases the downstream thermal resistance from 0.072 °C/W to 0.070 °C/W. In contrast, the rear-set baffle produces the opposite effect, raising the hotspot temperature to 77.05 °C, increasing the non-uniformity coefficient to 9.08, and elevating the downstream thermal resistance to 0.076 °C/W. The full-set baffle results in a milder deterioration, with corresponding values of 75.92 °C, 8.24, and 0.073 °C/W. A noteworthy observation is that the warmer downstream chip appears at CPU3 in both the baseline and front-set configurations, whereas it shifts to CPU4 under the rear-set and full-set baffle arrangements, which may be associated with flow-field

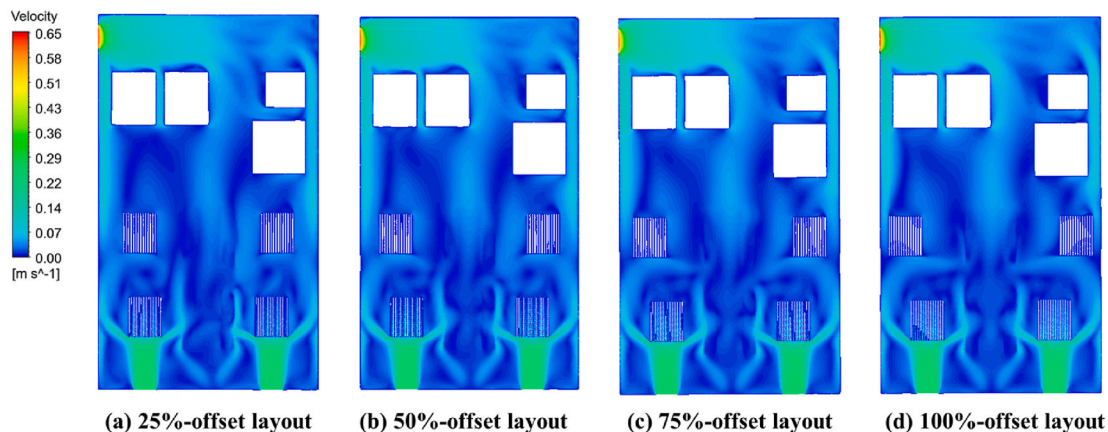


Fig. 9. Velocity fields under different lateral chip offset ratios ($X = 35$ mm).

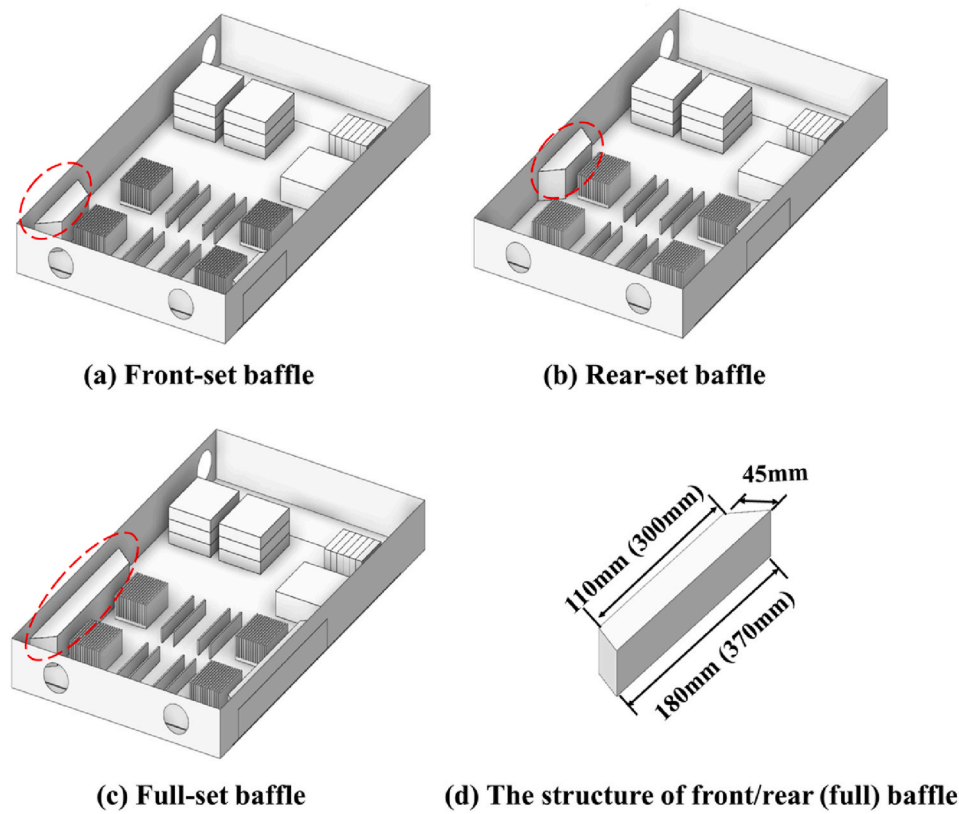


Fig. 10. Schematic of server layouts with different baffle configurations and the geometry of the baffle block.

Table 4

Thermal performance metrics of the server under different baffle configurations.

Baffle type	Baffle height (mm)	T_{CPU1} (°C)	T_{CPU2} (°C)	T_{CPU3} (°C)	T_{CPU4} (°C)	$T_{hotspot}$ (°C)	β	R_{rear} (°C/W)
Baseline	–	55.15	55.14	70.09	66.82	74.99	7.80	0.072
Front-set	60	54.92	54.91	68.47	65.59	73.36	7.09	0.070
	75	54.87	54.94	65.73	63.95	70.47	5.78	0.066
	90	54.94	54.90	62.14	63.40	66.71	4.56	0.063
Rear-set	60	55.10	55.11	69.12	72.23	77.05	9.08	0.076
	75	55.05	55.07	69.07	71.70	76.33	8.91	0.076
	90	54.94	54.98	70.35	65.81	74.67	7.80	0.072
Full-set	60	54.99	54.92	66.90	70.96	75.97	8.24	0.073
	75	54.88	54.89	63.19	66.06	70.62	5.74	0.066
	90	54.87	54.89	61.44	61.02	65.76	3.67	0.060

reorganization induced by the baffle arrangement.

Across all baffle layouts, increasing the baffle height leads to a systematic improvement in server thermal performance. Taking the full-set configuration as an illustrative example, raising the baffle height from 60 mm to 75 mm reduces the hotspot temperature from 75.97 °C to 70.62 °C, decreases the thermal non-uniformity coefficient from 8.24 to 5.74, and lowers the downstream thermal resistance from 0.073 °C/W to 0.060 °C/W. When the height further reaches 90 mm, the full-set baffle achieves the best thermal performance among all evaluated configurations, achieving a hotspot temperature of 65.76 °C, a non-uniformity coefficient of 3.67, and a thermal resistance of 0.060 °C/W. These values represent reductions of 9.23 °C, 4.13, and 0.012 °C/W relative to the baseline, respectively.

To interpret the performance differences reported above, the internal flow fields within the server are examined. As shown in Fig. 11, the front-set baffle suppresses the lateral dispersion of the bypass flow and forms a jet-like stream directed toward the downstream row. This concentrated cooling stream enhances local convective heat transfer, and thereby improves both temperature uniformity and inter-chip

thermal coupling. In contrast, the rear-set baffle redirects the bypass stream away from the downstream region, reducing the availability of low-temperature coolant and diminishing heat removal. The full-set baffle produces an intermediate response, combining the beneficial upstream guidance seen in the front-set case with the adverse downstream deflection characteristic of the rear-set layout. Importantly, the flow deflection induced by the rear-set and full-set baffles also weakens the obstruction imposed by the PSU on the right-side coolant pathway, mitigating thermal accumulation within the recirculation zone near CPU3. This flow-field restructuring accounts for the distinct downstream thermal behavior observed in these two configurations relative to the baseline.

Fig. 12 further illustrates the internal flow field under various baffle heights. Using the front-set baffle as an illustrative case, increasing the baffle height narrows the clearance between the baffle and the server top, thereby directing a larger fraction of the bypass coolant toward the downstream chips. flow confinement increases local coolant velocity and strengthens convective heat transfer. Compared with the front-set configuration, the full-set configuration provides a longer effective

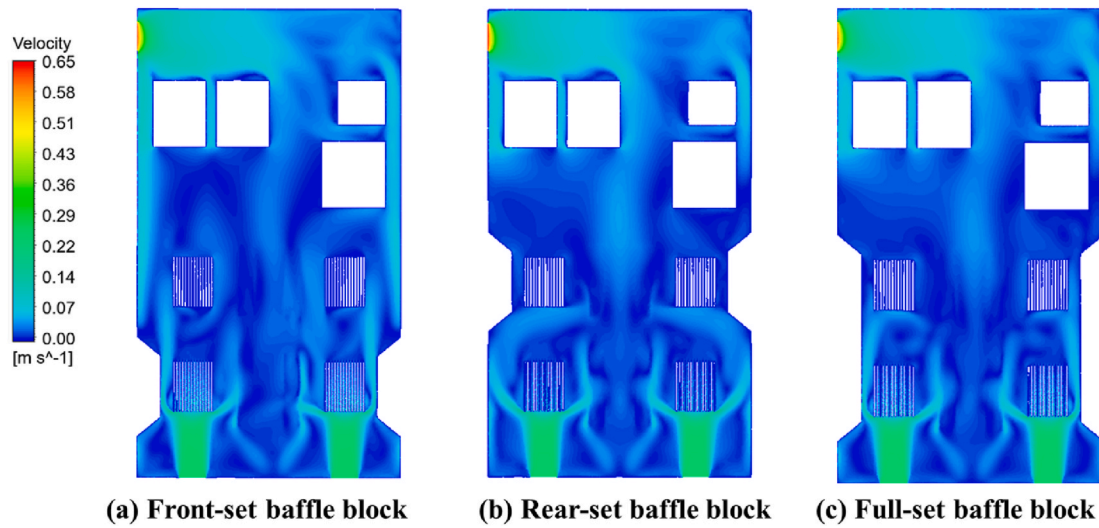


Fig. 11. Velocity fields inside the server under different baffle configurations ($X = 35$ mm).

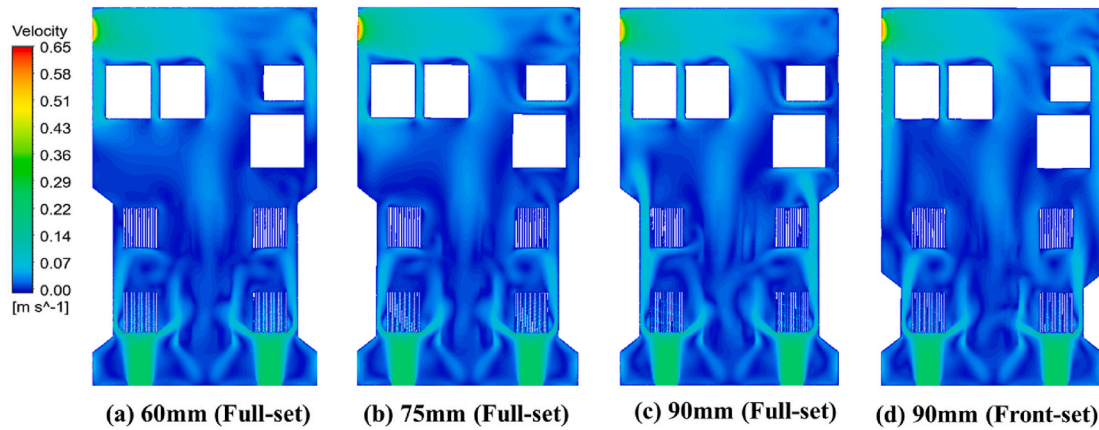


Fig. 12. Velocity fields inside the server under different baffle configurations ($X = 35$ mm).

guidance length at higher baffle heights, resulting in a broader jet-like region. Consequently, under the 90-mm condition, the full-set baffle delivers a more pronounced enhancement in heat transfer than the front-set case. From a hydraulic perspective, the installation of flow-guide baffles affects the pressure drop within the server. For the full-set configuration with a baffle height of 90 mm, the pressure drop is 369.67 Pa. The present section concentrates on the thermal impact of baffle-induced flow reorganization, while a detailed analysis of the pressure-drop mechanisms will be addressed in future work, which may be related to the fact that the primary flow passage is not directly obstructed by the baffles.

In summary, well-designed flow-guide baffle configurations provide a powerful means of reshaping coolant distribution and suppressing inter-chip thermal coupling in multi-chip servers. The baffle placement determines the dominant mode of flow-field reorganization, while the baffle height governs the strength and spatial extent of these reconstructed flow structures.

5. Conclusions

This study presented a multi-dimensional evaluation and optimization-oriented analysis of inter-chip thermal coupling in a single-phase immersion-cooled heterogeneous multi-chip server. By jointly considering chip temperature, thermal non-uniformity, and thermal resistance, the effects of coolant operating conditions and internal

structural layout on server thermal performance were quantitatively assessed. The main conclusions are summarized as follows.

- (i) Heterogeneous multi-chip servers exhibit pronounced upstream-downstream thermal coupling, increasing local hotspot thermal risk. Across the examined inlet-temperature range at an inlet velocity of 0.25 m/s, the downstream 600 W chip operates approximately 13.35 °C hotter than the upstream 900 W chip, indicating that chip thermal performance is determined by both coolant conditions and relative position along the flow path.
- (ii) Coolant inlet temperature and inlet velocity influence server thermal performance in different ways. Lowering the inlet temperature mainly reduces the overall temperature level and hotspot severity, with little impact on thermal coupling. Increasing inlet velocity improves thermal performance with diminishing returns, as upstream chips reach convective saturation first and determine the server-level critical velocity. Meanwhile, downstream chips remain more sensitive than upstream ones due to reheated coolant and lower local velocities.
- (iii) Structural design provides an effective lever to weaken thermal coupling by redistributing bypass coolant, yielding quantifiable thermal performance gains. Among all offset cases, a 75 % lateral chip offset achieves the best overall performance, with hotspot temperature, thermal non-uniformity coefficient, and downstream thermal resistance of 68.08 °C, 4.72, and 0.064 °C/W.

Well-arranged baffles reconstruct the flow field, with the baffle height governing the intensity and coverage of the redistributed flow. The 90 mm full-set baffle delivers the strongest enhancement, reducing hotspot temperature, non-uniformity coefficient, and downstream thermal resistance by 9.23 °C, 4.13, and 0.012 °C/W, respectively.

While the study clarifies the steady-state thermal behavior of immersion-cooled multi-chip servers and identifies effective structural and operational strategies, several limitations deserve further investigation. The present analysis relies primarily on single-parameter variations and does not fully capture the coupled influence of multiple operating and structural variables, nor the energy cost associated with optimized coolant delivery. Moreover, the steady-state framework omits the transient thermal response that arises under fluctuating workloads or during control adjustments. Future work will incorporate multi-parameter co-optimization, integrate thermal and energy consumption models, and evaluate server-level thermal performance under time-varying loads to develop thermal strategies that remain robust and energy-efficient in practical operation.

CRedit authorship contribution statement

Yang Lin: Writing – original draft. **Chang Yue:** Methodology. **Liping Zeng:** Investigation. **Yit Jing Ee:** Writing – review & editing. **Kamaruzzaman Sopian:** Writing – review & editing. **Wen Su:** Validation. **Jiaqiang Wang:** Conceptualization, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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